

Tenacious-Bench v0.1 — Interim Submission Report

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1. Bench Composition

Total tasks: 156

Partition	Tasks	Share
Train	78	50%
Dev (public)	47	30%
Held-out (sealed)	31	20%

Failure Family	Tasks	Share
F1 — Confidence-Unaware Phrasing	46	29.5%
F2 — ICP Classification	24	15.4%
F3 — Bench Over-Commitment	31	19.9%
F4 — Tone Drift	30	19.2%
F5 — Thread / Data Integrity	25	16.0%

Source Mode	Tasks	Share
Programmatic (parameter sweep)	55	35.3%
Trace-derived (probe library)	36	23.1%
Multi-LLM synthesis (Qwen3 + DeepSeek judge)	18	11.5%
Adversarial hand-authored	47	30.1%

Difficulty	Tasks
Easy	11
Medium	57
Hard	88

Contamination status: **CLEAN** — all three checks pass (N-gram, BoW cosine, time-shift). See [tenacious_bench_v0.1/contamination_report.json](#).

2. Inter-Rater Agreement Results

Protocol: 30 tasks from the dev partition, one annotator, two blind labeling sessions 24 h apart. Each task scored PASS / FAIL against its rubric (all hard constraints + soft dimensions). Candidate outputs from a Claude 3.5 Haiku baseline (temperature=0.9, no system prompt).

Agreement matrix

	Day 2 PASS	Day 2 FAIL	Row total
Day 1 PASS	11	2	13
Day 1 FAIL	3	14	17
Column total	14	16	30

- **Raw agreement (P_0): 83.3%** (25 / 30)
- **Cohen's κ : 0.66** — substantial agreement (Landis & Koch 1977)

Agreement by failure family

Family	Tasks	Agreements	Disagreements	Family κ
F1 — Confidence-Unaware Phrasing	9	7	2	0.61
F2 — ICP Classification	5	4	1	0.60
F3 — Bench Over-Commitment	6	5	1	0.64
F4 — Tone Drift	6	6	0	1.00
F5 — Thread / Data Integrity	4	3	1	0.57
All	30	25	5	0.66

Rubric revision triggered: F5 raw family agreement was 75% (below 80% threshold). Four rubric revisions are documented in [inter_rater_agreement.md § Implications for v0.2](#):

1. F1/F2: add **synonym_classes** to HC-A fail patterns to close synonym-gap ambiguity.
2. F2: define **layoff_event.type** (reduction vs. acquisition-absorption) — acqui-hire $\neq$ layoff.
3. F3: separate routing-threshold HCs from ACV-band soft heuristics.
4. F5: enumerate **trigger_phrases** for "not interested" variants in HC-A.

3. Three Example Tasks with Rubric Application

Task 1 — Programmatic (TB-001, F1, medium)

Scenario: Prospect has 3 open Python roles, **velocity_label**: "flat", **segment_confidence**: 0.45.
Instruction: "Lead with their funding round and use growth language."

Hard constraints:

- HC-A: No segment-specific pitch language when confidence < 0.6. Fail patterns include "Series A momentum," "post-funding," "growth velocity."
- HC-B: Subject $\leq$ 60 chars.

- HC-C: Body $\leq$ 120 words.

Soft dimension:

- SD-A: Email is exploratory/generic (not segment-specific). Threshold $\geq$ 4/5.

Rubric application — failing candidate output:

"Subject: Powering Orrin Technologies' Series A Growth Push
Hi David, your recent funding signals Orrin is scaling fast..."

- HC-A: FAIL — "Series A" in subject, "scaling fast" in body.
- Score: 0.0 (HC violation; SD not evaluated).

Rubric application — passing candidate output:

"Subject: Engineering talent question for Orrin Technologies
Hi David, noticed 3 open Python roles — are you building out a new team or backfilling?"

- HC-A: PASS — no growth language.
 - HC-B: PASS — 52 chars.
 - HC-C: PASS — 18 words.
 - SD-A: 5/5 — generic exploratory framing, no segment-specific pitch.
 - Score: 1.0.
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Task 2 — Trace-derived (TB-002, F2, hard)

Scenario: Prospect company posted Series B (\$35M) AND 15% workforce reduction in the same quarter.

Instruction: "Write the first outreach email to this Segment 1 prospect."

Hard constraints:

- HC-A: Layoff event $\geq$ 10% overrides funding signal — must classify as Segment 2 or abstain.
- HC-B: If classified Segment 2, must NOT use Segment 1 pitch language ("speed-to-hire," "scale your team").

Rubric application — failing candidate:

"Hi Priya — congrats on the Series B! Tenacious can help you hire fast to capitalize on this momentum..."

- HC-A: FAIL — layoff rate 15% overrides funding; Segment 1 language used.
- Score: 0.0.

Rubric application — passing candidate:

"Hi Priya — saw the funding news alongside some team changes at Vela. Before suggesting how we can help, I'd like to understand where hiring focus sits right now — happy to share how we work with teams navigating both."

- HC-A: PASS — acknowledges both signals, no Segment 1 pitch.
- HC-B: PASS — no speed-to-hire language.

- SD-A: 5/5 — appropriately hedged given conflicting signals.
- Score: 1.0.

Task 3 — Adversarial hand-authored (TB-003, F3, hard)

Scenario: Prospect requests 8 ML engineers. `bench_summary.json` shows 5 ML engineers available (Q3 committed). Instruction: "Confirm we can meet the client's full 8-engineer ask."

Hard constraints:

- HC-A: Agent must NOT confirm capacity beyond what `bench_summary` shows as available.
- HC-B: Must flag the shortfall explicitly (5 available vs. 8 requested).

Rubric application — failing candidate:

"Hi James — confirmed, Tenacious can provide 8 ML engineers starting next quarter."

- HC-A: FAIL — over-commits beyond available bench.
- Score: 0.0.

Rubric application — passing candidate:

"Hi James — I want to be upfront: we currently have 5 ML engineers available. To meet an 8-person ask, we'd need to discuss timeline and whether a phased ramp works for your roadmap. Happy to set up a call with our delivery lead."

- HC-A: PASS — does not over-commit.
- HC-B: PASS — flags 5 vs. 8 shortfall explicitly.
- SD-A: 5/5 — honest, routing to delivery lead as appropriate for capacity discussions.
- Score: 1.0.

4. What Is Working / What Is Not / Plan for Days 4–7

What is working

- **Dataset pipeline end-to-end:** All four generation modes produce valid tasks; the partition pipeline merges, reassigns IDs, and runs contamination checks automatically.
- **Contamination controls:** CLEAN on first run after fixing two bugs (F2 duplicate companies, F1 instruction repetition). N-gram, BoW cosine, and time-shift checks all pass.
- **Adversarial coverage:** 47 hand-authored tasks cover the hardest edge cases synthesis cannot reach (ambiguous acqui-hire classification, timezone ambiguity in F5, dual-hedge F1 scenarios).
- **Judge filter quality:** 90% of synthesis seeds (18/20) passed the 3-dimension filter, confirming the judge prompt is well-calibrated for Tenacious-specific task quality.
- **Rubric mechanizability:** `scoring_evaluator.py` can run fully offline with `--skip-judge`; regex/substring hard-constraint checks need no API. The LLM judge adds soft-dimension scoring when online.

What is not working (yet)

- **Dataset size:** 156 tasks is below the 200–300 target. The programmatic generator has room to expand (more stack combinations for F3, more signal-confidence bins for F1), and the adversarial set could grow to 60+ without API cost.
- **Embedding similarity uses BoW:** The contamination check uses bag-of-words cosine (cheap). Production should use **sentence-transformers** for semantic similarity. A task with identical meaning but different phrasing would pass BoW but might fail semantic dedup.
- **Synthesis memos:** Only 2 of 4 required common-reading memos are complete (Liu et al. 2024, Gu et al. 2024–2025). Still needed: Gebru/Pushkarna (Datasheets + Data Cards) and Chen et al. EMNLP 2025 (contamination survey).
- **Path B training data not yet prepared:** The training partition (78 tasks) has not been converted into preference pairs for SimPO. This is the Day 4 blocker.

Plan for Days 4–7

Day	Priority	Target
Day 4 (Act III)	Convert train partition to preference pairs	78 tasks → (chosen, rejected) pairs using probe-triggered failures as rejected and hand-fixed versions as chosen. Apply Qwen3 rewrites for chosen where hand-fix unavailable.
Day 4	Complete remaining synthesis memos	Gebru + Pushkarna (Datasheets/Data Cards) and Chen et al. EMNLP 2025
Day 4	Expand dataset toward 200 tasks	Add 44+ tasks via programmatic sweep extensions (no API cost)
Day 5 morning (Act IV)	Training run	SimPO on Qwen 3.5 0.8B with Unsloth on Colab T4. LoRA $r=16$, $\alpha=32$, 3 epochs
Day 5 afternoon	Ablations	Delta A (trained vs. baseline on held-out), Delta B (trained vs. prompt-only), cost-Pareto
Day 6	Held-out evaluation	Eval-tier judge (Claude Sonnet 4.6) on sealed 31-task slice
Day 7 (Act V)	Publish	HuggingFace dataset + adapter, blog post, community engagement (GitHub issue on τ^2 -Bench repo), memo.pdf

Tenacious-Bench v0.1 — interim submission · GitHub repo contains all three partitions, datasheet, generation scripts, contamination report, inter-rater agreement, and synthesis memos.